

PROFESSIONAL SUMMARY

Computer Engineer with over 5 years of automotive and defence Functional Safety (FuSa) experience. Comprehensive expertise in safety lifecycle management according to ISO 26262:2018, IEC 61508, and ISO 25119, encompassing Safety Case development with Goal Structuring Notation (GSN), Safety Plans, DIAs, and full requirements traceability aligned with ASPICE and OEM standards (SS 7740). Proven track record executing qualitative and quantitative safety analyses (FMEA, FTA, PHA, HAZOP), conducting safety-critical embedded software verification (MISRA, timing analysis, unit testing) on complex ECUs (OBC, SEooC, MICROSAR Safe), and representing safety teams during customer and OEM audits. Pioneer in applying human-in-the-loop Generative AI (RAG and LLM agents) to streamline FuSa compliance and analysis workflows.

CERTIFICATIONS & LANGUAGES

TÜV SÜD Certified: ISO 26262:2018 (Level 1)	English: C1 Advanced — EF SET (Score Verification)
TÜV SÜD Certified: IEC EN 61508:2010 (Level 1)	Italian: Native proficiency

PROFESSIONAL EXPERIENCE

Functional Safety Engineer | Kineton

Jun 2021 – Jul 2026 | Naples, Italy

Led automotive and defence Functional Safety engineering under ISO 26262 and ISO 25119, managing work products, embedded software verification, and process compliance aligned with ASPICE and OEM requirements via SS 7740.

- Safety Lifecycle & Team Coordination: Formulated, coordinated, and monitored comprehensive Functional Safety plans, milestones, and deliverables; mentored and guided project safety engineering team members across multi-disciplinary functions.
- Safety Cases & Documentation: Authored, structured, and maintained core FuSa artifacts including Safety Cases with Goal Structuring Notation (GSN) arguments, Safety Plans, Development Interface Agreements (DIA), confirmation measures, and ISO 26262 / ISO 25119 compliance checklists.
- Automotive & Defence Safety Analyses: Performed software FMEA and FTA on an On-Board Charger (OBC) control system board using specialized safety software; prepared Preliminary Hazard Lists (PHL) and executed PHA, SRHA, SSHA, and HAZOP for defence safety scopes.
- Embedded SW Verification & Architecture: Defined software architecture and integration requirements for safety-relevant components (SEooC, MICROSAR Safe, MCAL); conducted SW unit design, unit testing, and peer reviews, addressing MISRA compliance and static analysis findings.
- Timing & Real-Time Behavior: Instrumented and analyzed timing, execution latency, and runtime behavior of safety mechanisms and safety-critical execution paths to validate deterministic performance.
- Audits & Assessment Leadership: Served as primary technical contact for safety and process topics with customers and Tier-1 suppliers; successfully prepared and led teams through FuSa, ASPICE, and OEM assessments and audits.
- Generative AI Innovation: Pioneered human-in-the-loop Generative AI workflows (RAG pipelines and autonomous agents) to automate compliance verification, evidence synthesis, and traceability checking, significantly accelerating documentation cycles.

Automotive Software Intern | Kineton

Mar 2021 – Jun 2021 | Naples, Italy

- Embedded C/C++ Development: Developed firmware modules, build systems, and debugging tooling for automotive control units.
- Integration Requirements: Analyzed and managed system/software integration requirements to ensure baseline consistency.

TECHNICAL COMPETENCIES & DOMAIN EXPERTISE

Safety Standards & Governance ISO 26262:2018 (FuSa), IEC 61508 (Functional Safety), ISO 25119 (Agriculture & Machinery), Automotive SPICE (ASPICE), SS 7740:2023 (OEM process tailoring), Requirements Traceability, DIA, Safety Plans, Safety Cases (GSN), Confirmation Measures, Audits & Assessments.	Safety & Hazard Analyses Software FMEA & FTA (e.g., On-Board Charger - OBC ECU control board), Hazard Identification, Preliminary Hazard List (PHL), Preliminary Hazard Analysis (PHA), System/Subsystem Hazard Analysis (SRHA, SSHA), HAZOP, Risk Reduction Strategies.
Embedded Software & Verification C/C++, Python, SW Unit Design & Unit Testing, Peer Code Reviews, Code Inspections, MISRA C/C++ Compliance, Static Analysis Remediation, Real-Time Execution Path Profiling, Timing Analysis, SW Architecture & Integration Requirements.	Automotive Components & Tooling Safety Elements out of Context (SEooC), Vector MICROSAR Safe, Microcontroller Abstraction Layer (MCAL), Build Systems, Debugging Tooling, Git/GitHub, Online Safety Analysis Tools.
AI, Automation & Emerging Tech Generative AI applied to FuSa workflows (human-in-the-loop), RAG (Retrieval-Augmented Generation), Autonomous Compliance Agents, Microsoft Azure, Google Colab, Keras, TensorFlow, OpenFace.	Key Methodologies & Soft Skills Safety Planning & Coordination, Cross-Functional Team Mentorship, Customer & Supplier Technical Liaison, Technical Safety Documentation, Audit Presentation & Defense, Rigorous Engineering Traceability.

SELECTED TECHNICAL & SAFETY HIGHLIGHTS

On-Board Charger (OBC) Control ECU:	Executed exhaustive software FMEA and FTA on an OBC control board, mapping safety mechanisms to diagnostic requirements and hardware fault metrics.
Defence Safety Verification:	Formulated Preliminary Hazard Lists (PHL) and carried out PHA, SRHA, SSHA, and HAZOP methodologies for mission-critical defence systems.
Embedded Architecture & SEooC:	Integrated Safety Elements out of Context (SEooC) within AUTOSAR / Vector MICROSAR Safe and configured MCAL hardware abstraction layers.
GenAI in Safety Compliance:	Designed human-in-the-loop RAG and agent architectures to accelerate safety standard cross-referencing, evidence gathering, and checklist verification.

EDUCATION & ACADEMIC BACKGROUND

Master’s Degree in Computer Engineering University of Naples Federico II	Sep 2015 – Mar 2021 Naples, Italy
Final Grade: 107 / 110 • Thesis: Multimodal Analysis of Emotions in Multimedia. • Implementation: Designed and deployed an end-to-end Emotion Recognition machine learning application leveraging Microsoft Azure and Google Colab cloud platforms.	
Erasmus+ Traineeship — Machine Learning & Computer Vision Universidad Politécnica de Madrid (ETSISI)	Sep 2020 – Feb 2021 Madrid, Spain
Final Grade: Sobresaliente (10 / 10 with distinction) • Applied R&D: Developed a high-precision computer vision pipeline for facial emotion recognition utilizing Python, Keras, TensorFlow, OpenFace, Azure, and Git.	
Bachelor’s Degree in Computer Engineering University of Naples Federico II	Sep 2009 – Jan 2015 Naples, Italy
Final Grade: 96 / 110 • Thesis: NoSQL Technologies: Sesame. • Implementation: Conducted comparative analysis of graph database architectures with practical focus on the Sesame RDF database manager implementation.	

HONORS & ADDITIONAL EXPERIENCE

- Academic Merit Scholarship Recipient (2009 – 2012): Awarded competitive university scholarship by University of Naples Federico II based on sustained academic standing and merit.
- Community Volunteer — Freebacoli (Sep 2010 – May 2015): Actively contributed to local non-profit community engagement, youth mentorship, and civic transparency initiatives in Bacoli (Naples).

In compliance with the Italian Legislative Decree no. 196 dated 30/06/2003 and the EU GDPR 2016/679, I hereby authorize the processing of my personal data for recruitment and selection purposes.